

## **Group Project Human-Computer Interaction: Group Project Proposal**

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## Introduction

The lack of access to information related to mental well-being continues to pose a considerable problem to many people due to factors such as financial constraints, stigma, and the limited availability of professional support. Chatbot technologies have gained relevance due to their capacity to provide timely assistance through conversational interactions with users (Minaee et al., 2021). However, general-purpose chatbots may generate erroneous or fabricated responses, which can be particularly problematic in healthcare contexts. The proposed application will address these concerns by incorporating a Retrieval-Augmented Generation (RAG) approach that uses WHO guidelines and CBT-related resources to generate responses, thereby reducing the likelihood of misinformation (Lewis et al., 2020; World Health Organization, 2025). Apart from functional considerations, the application will incorporate Human-Computer Interaction (HCI) principles to ensure that the user interface is user-friendly, accessible, and appropriate for its intended audience. The chatbot is intended solely for educational and informational purposes and is not designed to diagnose, treat, or replace professional mental health services.

## Proposed Project Scope

The proposed project aims at creating a Retrieval-Augmented Generation (RAG)-based Mental Health Support Chatbot that will facilitate giving user credible answers to frequent mental health and wellness queries. The chatbot will access reliable publicly available resources like World Health Organization (WHO) manuals on mental health and cognitive behavioral therapy (CBT) materials and produce answers (World Health Organization, 2025). The system will be built to enhance ease of access to information about mental health via a conversational and user-friendly interface.

Python and open-source technologies based on artificial intelligence will be used to develop the project. The interface of chatbot is likely to be created with Gradio and implemented on Hugging Face Spaces or Google Colab to be easily available (Hugging Face, 2026) and testable. The system will emphasize educational and informational assistance, as opposed to professional medical diagnosis or treatment. Project scope includes the development of a chatbot, the implementation of a retrieval system, interface design, testing, evaluation of the quality of user interaction, and response.

## Project Objectives

The main goal of the project is to develop and design a conversational AI chatbot capable of responding with relevant and precise answers to mental health issues. To help avoid hallucinations and enhance reliability, the chatbot will apply Retrieval-Augmented Generation (RAG) methods to retrieve information on trusted sources prior to answer generation (Lewis et al., 2020).

The other objective is to argue about the principles of Human-Computer Interaction (HCI) in conversational AI systems and create a convenient, friendly and simple to use interface (Gradio Team, 2025). An additional aim of the project is to demonstrate how technologies of artificial intelligence, natural language processing, and information retrieval can be effectively applied in real-life problems related to the support of mental wellness and its accessibility.

## Technologies and Tools

The proposed system will be developed in Python using a Retrieval-Augmented Generation (RAG) architecture that combines document retrieval and answer generation within a unified workflow (Lewis et al., 2020). The application will be designed to ingest PDF documents using

the pypdf library and automatically divide them into smaller text passages to improve retrieval precision and contextual relevance.

To enable semantic search, each text passage will be converted into a vector embedding using the all-MiniLM-L6-v2 model from the sentence-transformers framework. These embeddings will be stored in a FAISS vector database, allowing the system to rapidly identify document passages that are most relevant to a user's query (Minaee et al., 2021).

When a question is submitted, the system will retrieve the highest-matching passages from the vector index and provide them as contextual input to a large language model (Vaswani et al., 2017). Gemini 2.5 Flash is planned as the primary response-generation model due to its speed and reasoning capabilities. To provide flexibility and support comparative evaluation, Groq's Llama 3.3 70B and a locally deployed Flan-T5 model will also be integrated as alternative response engines.

The user interface will be developed using Gradio, providing a web-based environment through which users can upload documents and interact with the chatbot (Gradio Team, 2025). The completed application will be deployed on Hugging Face Spaces to enable cloud-based access and demonstration. GitHub will serve as the project's source code repository, supporting version control, issue tracking, and collaborative development among team members. GitHub Actions may also be used to create a simple CI/CD workflow that automatically builds and tests the chatbot when code changes are pushed (GitHub, n.d). This would help confirm that required Python packages install correctly, that the retrieval pipeline works as expected, and that updates to the chatbot interface or response-generation logic do not introduce new errors.

A key feature of the proposed solution is its ability to generate responses grounded in uploaded documents rather than relying solely on the language model's pre-trained knowledge. This approach is expected to improve answer accuracy, reduce hallucinations, and provide users with responses that are directly tied to the source material. Table 1 summarizes the technologies selected for the project, their intended roles, and their corresponding access locations.

| Role in the project                | Tool                                     | Purpose                                                                 | Resource                                                                  |
|------------------------------------|------------------------------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Code environment                   | Google Colab                             | Browser-based Python with free GPU/CPU; nothing to install              | colab.research.google.com                                                 |
| Document reading                   | pypdf                                    | Extracts text from PDF guides                                           | pypi.org/project/pypdf                                                    |
| Text → embeddings                  | sentence-transformers (all-MiniLM-L6-v2) | Converts text into vectors locally, no API key                          | <a href="https://www.sbert.net/">https://www.sbert.net/</a>               |
| Vector search                      | FAISS (faiss-cpu)                        | Stores and searches embeddings                                          | <a href="https://faiss.ai/">https://faiss.ai/</a>                         |
| Answer generation                  | Google AI Studio – Gemini 2.5 Flash      | Generates grounded answers (primary model)                              | <a href="https://aistudio.google.com/">https://aistudio.google.com/</a>   |
| Generation (alternative)           | Groq – Llama 3.3 70B                     | Fast fallback model                                                     | <a href="https://console.groq.com/">https://console.groq.com/</a>         |
| Generation (offline)               | Flan-T5 (Hugging Face transformers)      | Fully local, no-API-key option                                          | <a href="https://huggingface.co/">https://huggingface.co/</a>             |
| Web interface                      | Gradio                                   | Builds the chat UI in a few lines of code                               | <a href="https://www.gradio.app/">https://www.gradio.app/</a>             |
| Hosting / sharing                  | Hugging Face Spaces                      | Free public hosting and demo URL                                        | <a href="https://huggingface.co/spaces">https://huggingface.co/spaces</a> |
| Source code, versioning, and CI/CD | GitHub and GitHub Actions                | Stores code, supports collaboration, and automates build/test workflows | <a href="https://github.com/">https://github.com/</a>                     |

*Table 1. Project technology stack.*

## System Overview

The proposed system is a Retrieval-Augmented Generation (RAG)-based mental health chatbot designed to provide informative and context-aware responses. Users interact with the system through a simple Gradio-based web interface by entering mental health-related questions.

Once a query is submitted, the system processes the input and converts it into vector embeddings using a sentence-transformer model. These embeddings are then compared against a pre-built FAISS vector database containing chunks of curated mental health documents, such as WHO and CBT resources to identify the most relevant information for a user's query (Minaee et al., 2021).

The most relevant text passages are retrieved and passed to a language generation model. Gemini 2.5 Flash is planned as the primary model for generating responses because of its speed and strong reasoning capabilities. Groq's Llama 3.3 70B and a locally deployed Flan-T5 model may also be used as alternative response engines for comparison, testing, or fallback purposes. The selected model then generates a final response grounded in the retrieved information to improve accuracy and reduce hallucinations (Vaswani et al., 2017).

## Expected Features

The proposed mental health chatbot will include several features designed to improve usability, response accuracy, accessibility, transparency, and overall user experience. The main feature will be a conversational question-and-answer interface where users can ask mental health and wellness-related questions in plain language. The chatbot will use Retrieval-Augmented Generation (RAG) to retrieve relevant passages from trusted mental health resources before generating a response. This will help ensure that the answers are grounded in source material rather than relying only on the language model's pre-trained knowledge (Lewis et al., 2020). The

system will also use semantic search through vector embeddings and FAISS so that user questions can be matched with the most relevant sections of WHO and CBT-related documents (Minaee et al., 2021; World Health Organization, 2025).

From an HCI perspective, the chatbot will be designed with a simple Gradio-based web interface that emphasizes readability, ease of use, and accessibility (Gradio Team, 2025). The system may also include source-aware responses, allowing users to see or understand what retrieved information influenced the chatbot's answer. This feature would improve transparency and help users develop appropriate trust in the system. In addition, the chatbot will support multiple language models, including Gemini 2.5 Flash, Llama 3.3 70B, and Flan-T5, so the team can compare response quality, speed, and reliability. Since modern language models are largely based on transformer-based architectures, this comparison also supports the project's broader exploration of natural language generation tools (Vaswani et al., 2017).

Because the project involves mental health information, the chatbot will include an educational-use disclaimer explaining that it does not provide diagnosis, treatment, emergency support, or professional medical advice. The system should also include basic safety handling for sensitive questions by providing cautious, informational responses and encouraging users to seek professional help when necessary. Finally, the completed application will be deployed through Hugging Face Spaces for public access and demonstration, with GitHub supporting version control, collaboration, and possible CI/CD build and testing workflows (GitHub, n.d.; Hugging Face, 2026).

## Group Collaboration and Work Distribution

A collaborative approach will be applied in which team members will participate at different stages of the project. This includes stages like research, programming, testing,

documenting, project designing, and preparation of the presentation based on their past experiences, strengths, and preferences. While some team members may become more active at technical implementations including coding, testing, debugging, and system integration, other team members may participate actively in documenting the work, project design, conducting research, organizing, and preparing presentations. This will ensure that the project will benefit from various technical and analytical strengths of team members while also ensuring a collaborative effort during the project period.

## Anticipated Challenges

Several technical and practical challenges are anticipated during the development of the project. One major challenge involves ensuring the accuracy and reliability of generated responses. Although Retrieval-Augmented Generation (RAG) reduces hallucinations by grounding responses in retrieved documents (Lewis et al., 2020), there is still a possibility that the chatbot may generate incomplete, unclear, or contextually weak answers. Careful selection of trusted data sources and testing of retrieval performance will therefore be necessary.

Another challenge involves data preparation and document management. The quality of chatbot responses depends heavily on the quality and organization of the mental health resources used within the retrieval database. Preparing documents, cleaning text data, splitting content into searchable sections, and generating embeddings may require additional time and experimentation.

In addition, designing a conversational interface that follows Human-Computer Interaction (HCI) principles may present usability challenges. The chatbot should provide clear, understandable, and supportive responses while maintaining a user-friendly and accessible



interface. Testing and improving user experience will therefore be an important part of the project.

Finally, ethical considerations and responsible AI usage are important challenges for a mental health-related system. The chatbot is intended only for informational and educational purposes and not for professional diagnosis or treatment. Appropriate disclaimers and limitations must be included to ensure users understand the boundaries of the system and avoid overreliance on AI-generated responses.

## Conclusion

To sum up, the suggested RAG-based Mental Health Support Chatbot will merge the principles of artificial intelligence and human-computer interaction to deliver a convenient and trustworthy source of mental health information in the form of a conversational exchange. With the help of trusted sources and response generation based on retrieval, the system aims to enhance the accuracy of the responses and minimize risks of misinformation and hallucinations that are typical of generative AI systems (Lewis et al., 2020).

Another important learning experience that will be gained during the project is the development of AI, chatbot design, natural language processing, retrieval systems, and team software development (Lewis et al., 2020; Minaee et al., 2021). Altogether, the suggested system is a feasible and realistic AI application that can be used to fulfill the goals of the course on Human-Computer Interaction.

## AI Use Disclosure

Artificial intelligence tools including ChatGPT, Claude, and Grok, were used during the planning and brainstorming stages of this proposal to assist with organization, idea development, and support. All content was reviewed, revised, and verified by the team prior to submission.

**Project Repository:** <https://github.com/Dr0m0x/MSAI631-mental-health-rag-chatbot>

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